

Theory section

Introduction

I_b - Design current

I_n - Protective size

I_z - current-carrying capacity (with all corrections)

I_t - current-carrying capacity.

$$Z_s = Z_e + (R_1 + R_2)$$

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Z_s = Earth Fault Loop Impedance (EFLI)

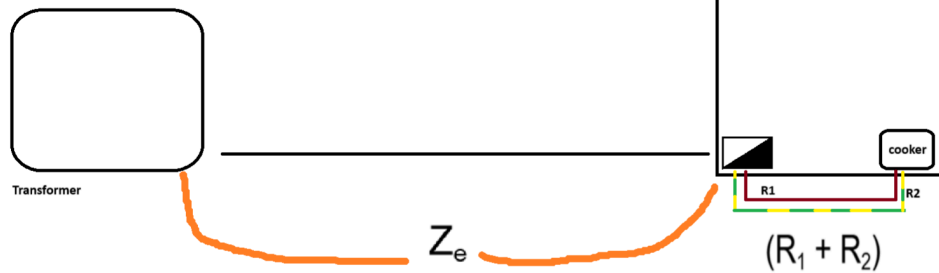
Z_e = Connection from fuse box back to transformer

$(R_1 + R_2)$ = Line + CPC resistance. Internal from the circuit to the fuse box

R_1 = Live

R_2 = Earth

Z = Impedance



Question:

A 230 V 4kW load is to be wired in non-armoured single-core 70°C thermoplastic insulated copper conductors wired in steel conduit installed on the surface some 25 metres from the distribution board.

Three other similar circuits are installed in the same conduit which passes through an area where the ambient temperature reaches 40°C.

Calculate the cross-sectional area of cable required if protection is by a BS288-2 fuse. Show all working.

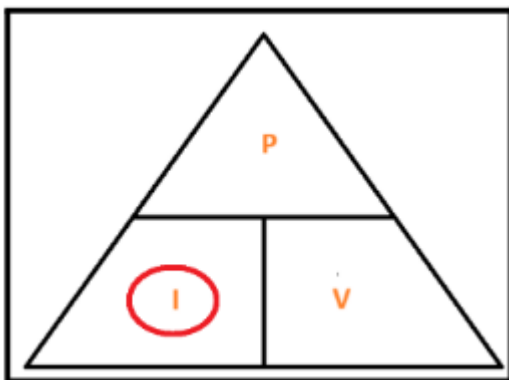
Working out:

Voltage (V) - informs us what kind of circuit or power supply is needed

Kilowatts (kW) - informs us of how much power is consumed or required

(Used ChatGPT to get these definitions)

1) $I_b \rightarrow$ design current



$$I = P/V$$

$$4,000/230 = 17.39 \text{ A}$$

We do not calculate the diversity rule because the full load is not specified.

$$I_b = 17.39 \text{ A}$$

2) I_n -> Protective device

$$I_n = 20 \text{ A}$$

3) I_t - Current carrying capacity

$$I_z \Rightarrow I_t \geq \frac{I_n}{C_a \times C_g \times C_i \times C_f} \Rightarrow \frac{I_n}{C_a \times C_g}$$

C_a => Ambient temperature.

$$I_z \Rightarrow I_t \geq \frac{I_n}{C_a \times C_g \times C_i \times C_f} \Rightarrow \frac{I_n}{0.87 \times C_g}$$

The ambient temperature is defined as the air temperature surrounding a particular object or area. The question states that "the ambient temperature reaches 40 °C". The questions also states "thermoplastic 70°C insulated copper conductors"

	<u>Insulation</u>			
			<u>Mineral</u>	
<u>Ambient temperature</u>	<u>70°C thermoplastic</u>	<u>90°C thermosetting</u>	<u>Thermoplastic covered or bare and exposed to touch 70°C thermoplastic</u>	<u>cBare and not exposed to touch 105 °C</u>
<u>25</u>	<u>1.03</u>	<u>1.02</u>	<u>1.07</u>	<u>1.04</u>
<u>30</u>	<u>1.00</u>	<u>1.00</u>	<u>1.00</u>	<u>1.00</u>
<u>35</u>	<u>0.94</u>	<u>0.96</u>	<u>0.93</u>	<u>0.96</u>
<u>40</u>	<u>0.87</u>	<u>0.91</u>	<u>0.85</u>	<u>0.92</u>

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	<u>Insulation</u>			
			<u>Mineral</u>	
<u>Ambient temperature</u>	<u>70°C thermoplastic</u>	<u>90°C thermosetting</u>	<u>Thermoplastic covered or bare and exposed to touch 70°C thermoplastic</u>	<u>cBare and not exposed to touch 105 °C</u>
<u>25</u>	<u>1.03</u>	<u>1.02</u>	<u>1.07</u>	<u>1.04</u>
<u>30</u>	<u>1.00</u>	<u>1.00</u>	<u>1.00</u>	<u>1.00</u>
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Therefore, using the table the answer for C_a is 0.87.

$C_g \Rightarrow$ grouping

“Three other similar circuits are installed in the same conduit ...”

<u>Number of circuits or multi-core cables</u>											
<u>Arrangement (cable touching)</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>5</u>	<u>6</u>	<u>7</u>	<u>8</u>	<u>9</u>	<u>12</u>	<u>Applicable reference method for current-carrying capacities</u>
Bunched in air, on a surface, embedded or enclosed	1.0	0.80	0.70	0.65	0.60	0.57	0.54	0.52	0.50	0.45	A to F
Single layer on wall or floor	1.0	0.85	0.79	0.75	0.73	0.72	0.72	0.71	0.70	0.7	C
Single layer multicore on a perforated horizontal or vertical cable tray system	1.0	0.88	0.82	0.77	0.75	0.73	0.73	0.72	0.72	0.72	E
Single layer multicore on a cable ladder system or cleats etc	1.0	0.87	0.82	0.80	0.80	0.79	0.79	0.78	0.78	0.78	E

<u>Number of circuits or multi-core cables</u>											
<u>Arrangement (cable touching)</u>	<u>1</u>	<u>2</u>	<u>3</u>	<u>4</u>	<u>5</u>	<u>6</u>	<u>7</u>	<u>8</u>	<u>9</u>	<u>12</u>	<u>Applicable reference method for current-carrying capacities</u>
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Single layer multicore on a cable ladder system or cleats etc	1.0	0.87	0.82	0.80	0.80	0.79	0.79	0.78	0.78	0.78	E

$I_t = 37.37 \text{ A}$ (Current carrying capacity without corrections)

$$I_t \geq \frac{I_n}{C_a \times C_g \times C_i \times C_f} \Rightarrow \frac{20 \text{ A}}{0.87 \times 0.65} = 35.37 \text{ A}$$

When you see the voltages 230V or 400v just remember:

- 230V - single phase systems
- 400V - three phase systems.

Conductor cross-sectional area	Reference method A (enclosed in conduit in thermally insulating wall, etc.)		Reference method B (enclosed in conduit on a wall or in trunking, etc.)		Reference method C (clipped direct)		Reference method F (in free air or on a perforated cable tray horizontal or vertical)					
	2 cables, single-phase AC or DC	3 or 4 cables, three-phase AC	2 cables, single-phase AC or DC	3 or 4 cables, three-phase AC	2 cables, single-phase AC or DC flat and touching	3 or 4 cables, three-phase AC flat and touching or trefoil	Touching			Spaced by one cable diameter		
							2 cables, single-phase AC or DC flat	3 cables, three-phase AC flat	3 cables, three-phase AC trefoil	2 cables, single-phase AC or DC	3 cables, three-phase AC flat	
										horizontal	vertical	
1 mm ²	2	3	4	5	6	7	8	9	10	11	12	
1	11	10.5	13.5	12	15.5	14						
1.5	14.5	13.5	17.5	15.5	20	18						
2.5	20	18	24	21	27	25						
4	26	24	32	28	37	33						
6	34	31	41	36	47	43						

Table F4(i) - page 171

We used the above table to deduce the I_z value.

Remember that we deduced I_t to be **35.37 A**.

If we round this up, 41 is the next number (circled in red above)

The question states **230V** - which means it is a **single phase system**

The question states "enclosed into the same conduit" "...installed onto the surface".

The answer highlighted in **green**, shows that we need a **6mm²** cable size. **6mm²** cable corresponds to a **41 A** which is enough.

$I_z = 6 \text{ mm}^2$

Voltage drop

▼ Table F4(ii)

Table 4D18

Voltage drop (per ampere per metre) at a conductor operating temperature of 70 °C

Conductor cross-sectional area	2 cables DC	2 cables, single-phase AC	3 or 4 cables, three-phase AC				
	Reference methods A & B (enclosed in conduit or trunking)	Reference methods C & F (clipped direct on tray or in free air) touching	Reference methods A & B (enclosed in conduit or trunking)	Reference methods C & F (clipped direct, on tray or in free air) spaced	Reference methods C & F (clipped direct, on tray or in free air) Touching, Trefoil	Reference methods C & F (clipped direct, on tray or in free air) Touching, Flat	Reference methods C & F (clipped direct, on tray or in free air) Spaced*, Flat
mm ²	mV/A/m	mV/A/m	mV/A/m	mV/A/m	mV/A/m	mV/A/m	mV/A/m
1	44	44	44	44	38	38	38
1.5	29	29	29	29	25	25	25
2.5	18	18	18	18	15	15	15
4	11	11	11	11	9.5	9.5	9.5
6	7.3	7.3	7.3	7.3	6.4	6.4	6.4
10	4.4	4.4	4.4	4.4	3.8	3.8	3.8
16	2.8	2.8	2.8	2.8	2.4	2.4	2.4
25	1.75	1.80	1.75	1.80	1.55	1.50	1.55
35	1.25	1.30	1.25	1.30	1.10	1.10	1.15
50	0.93	1.00	0.95	0.97	0.85	0.82	0.86
70	0.63	0.72	0.66	0.69	0.61	0.57	0.60
95	0.46	0.56	0.50	0.54	0.48	0.43	0.47

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* Spacings larger than one cable diameter will result in larger voltage drop.

The impedance values in Table F4(ii) consist of both the resistive and reactive elements of voltage drop, usually provided separately for 25 mm² and above.

Appendix

Table F4(ii) page 173

$$\text{Volt Drop (Vd)} = \frac{\langle \text{mV/A/m} \rangle \times Ib \times L}{1000}$$

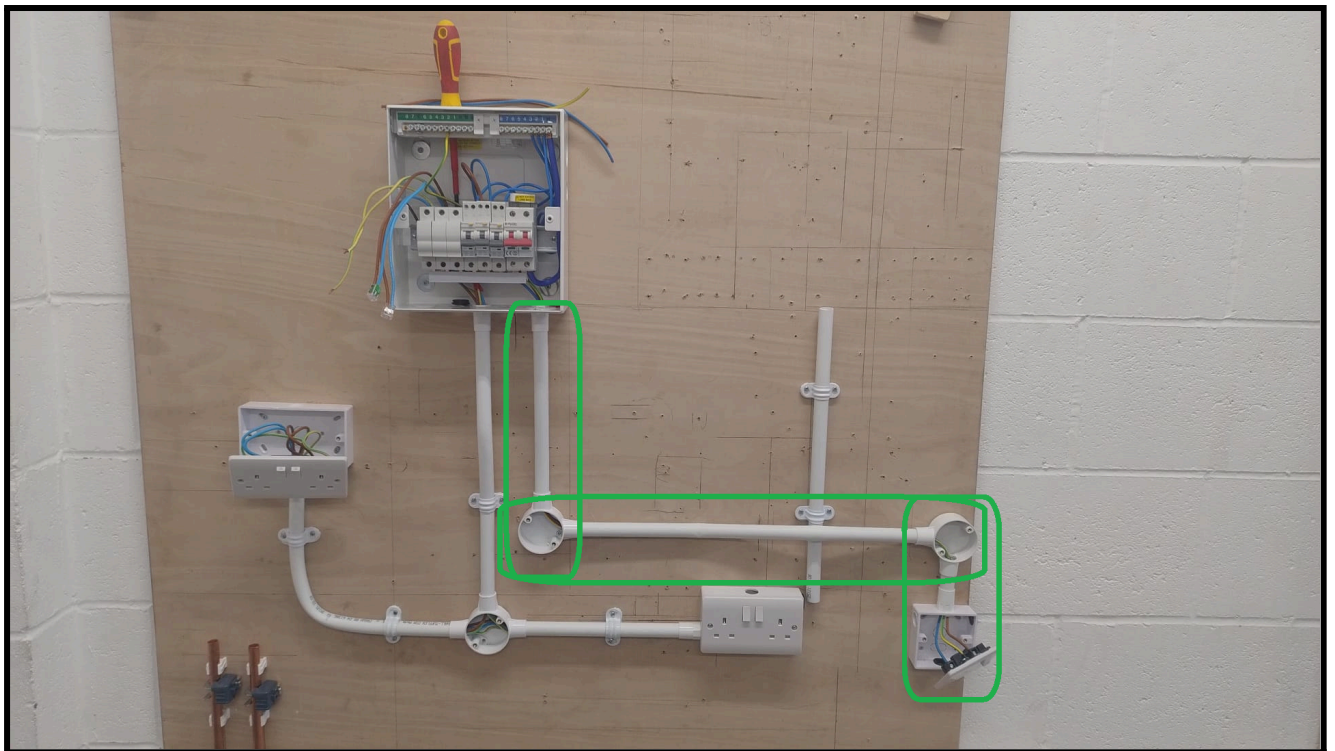
$$\text{Voltage Drop} = \frac{7.3 \times 17.39 \times 25}{1000} = 3.17 \text{ V}$$

Practical section

✳ Important Clarification:

Display	What it Means	Continuity?	
0.5 Ω	Very low resistance (good path)	✓ Yes	
50 Ω	Some resistance, might still beep	⚠ Maybe (depends)	
>200 Ω / OL	Too much resistance (open circuit)	✗ No	

In the board below we also installed a radial circuit - which is highlighted in green.



[illegible]

Test for a Radial circuit

Step One - test for Continuity (Ω)

$$(R_1 + R_2)$$

$R_1 = \text{Live}$ and $R_2 = \text{Earth}$



A) Set-up electrical tester for continuity test:

- Set the dial on the electrical tester to the “**continuity**” setting;
- Plug the tester socket into the plug;
- Attach the probes into the **Live** and **Earth** socket pins.

B) In terms of the Fuse board:

- Put the Earth wire into position two of the connector block;
- This is because the circuit that we wish to test is on the second RCD;
- The live wire can go anywhere in the connector block.

As a result of following **part A** and **part B** we create a loop in the circuit. A loop in which the electrical current flows down the **brown live wire** and back through the **green and yellow earth wire**.

C) Hit the test button on the electrical tester - interpret the result:

- First reading was **0.05 Ω (Ohms)**
- Second reading of **0.07 Ω (Ohms)**.
- This means that there is very little resistance in the circuit.
- When there is little resistance this means there is a good flow of current.
- Continuity checks if there is a connection between two points.



>200 Ω or there around - means no continuity or an open circuit. Resistance is too high to even measure.

Insulation resistance

Step Two - Test voltage (V)

Set the test voltage to 500 on the machine.

Write 500 on the test sheet because this is the voltage level.

Step Three - Live - Live (MΩ)

Means we test the **brown live wire** and the **blue neutral wire**.

We scored > 200Ω Ohms.

This test means that the **brown live wire** and the **blue neutral wire** do not touch each other in the radial circuit.

Does this score mean that because there is so much resistance the wires (**brown live wire** and the **blue neutral wire**) simply cannot be touching. Because if they were touching the resistance would be lower as there would be more current flow.

Step four - Live - Earth (MΩ)

Test the brown live wire and the earth wire. We got >200 Ω (Ohms) which means the wires are not touching.

Step five - Polarity is ticked because we got a reading in Part One subsection D.